

Algebra (7 th std Term 2)

Choose the appropriate answer

1. $a \times a \times a \times a \times a$ is equal to
(a) a^5 (b) $5 + a$ (c) $5a$ (d) $a + 5$

Ans: (a) a^5

2. The exponential form of 72 is
(a) 7^2 (b) 2^7 (c) $2^3 \times 3^3$ (d) $2^3 \times 3^2$

Ans: (d) $2^3 \times 3^2$

3. The value of x in the equation $a^{13} = x^3 \times a^{10}$ is
(a) a (b) 13 (c) 3 (d) 10

Ans: (a) a

4. How many zeroes are there in 100^{10} ?
(a) 2 (b) 3 (c) 10 (d) 20

Ans: (d) 20

$$100^{10} = (10^2)^{10} = 10^{20}$$

5. $2^{40} + 2^{40}$ is equal to
(a) 4^{40} (b) 2^{80} (c) 2^{41} (d) 4^{80}

Ans: (b) 2^{80}

6. Observe the equation $(10 + y)^4 = 50625$ and find the value of y.
(a) 1 (b) 5 (c) 4 (d) 0

Ans: (b) 5

7. The unit digit of $(32 \times 65)^0$ is
(a) 2 (b) 5 (c) 0 (d) 1

Ans: (d) 1

8. The unit digit of the numeric expression $10^{71} + 10^{72} + 10^{73}$ is
(a) 0 (b) 3 (c) 1 (d) 2

Ans: (a) 0

$$10^{71} + 10^{72} + 10^{73} = 0 + 0 + 0 = 0$$

9. $3p^2 - 5pq + 2q^2 + 6pq - q^2 + pq$ is a
(a) Monomial (b) Binomial (c) Trinomial (d) Quadrinomial

Ans: (c) Trinomial

10. The degree of $6x^7 - 7x^3 + 4$ is
(a) 7 (b) 3 (c) 6 (d) 4

Ans: (a) 7

11. If $p(x)$ and $q(x)$ are two expressions of degree 3, then the degree of $p(x) + q(x)$ is
(a) 6 (b) 0 (c) 3 (d) Undefined

Ans: (c) 3

12. The exponential form of 10000 is
(a) 10^3 (b) 10^4 (c) 10^5 (d) None of these

Ans: (b) 10^4

13. The exponential form of 100000 is
(a) 10^3 (b) 10^4 (c) 10^5 (d) None of these

Ans: (c) 10^5

14. The exponential form of 81 is
(a) 3^4 (b) 3^3 (c) 3^2 (d) None of these

Ans: (a) 3^4

15. The exponential form of 125 is
(a) 5^4 (b) 5^3 (c) 5^2 (d) None of these

Ans: (b) 5^3

16. The exponential form of 32 is
(a) 2^3 (b) 2^4 (c) 2^5 (d) None of these

Ans: (c) 2^5

17. The exponential form of 64 is
(a) 2^5 (b) 2^6 (c) 2^7 (d) 2^8

Ans: (b) 2^6

18. The value of $(-2)^3$ is
(a) 8 (b) -8 (c) 16 (d) -16

Ans: (b) -8

19. The value of $(-2)^4$ is
(a) 8 (b) -8 (c) 16 (d) -16

Ans: (c) 16

20. If $2^3 \times 2^4 = 2^?$, Then? =
(a) 3 (b) 4 (c) 1 (d) 7

Ans: (d) 7

21. If $(-3)^4 \times (-3)^6 = (-3)^?$, Then? =
(a) 4 (b) -3^{10} (c) -3^6 (d) -3^{12}

Ans: (b) -3^{10}

22. $2^7 \div 2^3 =$
(a) 2^4 (b) 2^{10} (c) 2 (d) $1/2$

Ans: (a) 2^4

23. $10^6 \div 10^5 =$
(a) 10^1 (b) 10^5 (c) 10^6 (d) 10^{11}

Ans: (a) 10^1

24. Simplify the expression $(-5)^4 =$
(a) 125 (b) 625 (c) 375 (d) 125

Ans: (b) 625

25. Simplify the expression $(2^2)^3 =$
(a) 2^2 (b) 2^3 (c) 2^1 (d) 2^6

Ans: (d) 2^6

26. Simplify the expression $(5^2)^{10} =$
(a) 52 (b) 5^{20} (c) 5^{10} (d) 5^5

Ans: (b) 5^{20}

27. $3^0 \times 4^0 \times 5^0 =$
(a) 1 (b) 3 (c) 4 (d) 5

Ans: (a) 1

28. $(2^0 + 3^0) \times 4^0 =$
(a) 1 (b) 2 (c) 3 (d) 4

Ans: (b) 2

29. Which of the following is true?
(a) $2^0 = (100)^0$ (b) $10^2 \times 10^8 = 10^{16}$ (c) $2^2 \times 3^3 = 6^5$ (d) $2^3 > 3^2$

Ans: (a) $2^0 = (100)^0$

30. Simplify the expression $(2^2 \times 2)^2 =$
(a) 2^3 (b) 2^4 (c) 2^5 (d) 2^6

Ans: (d) 2^6

31. $3^8 / (3^5 \times 3^3) =$
(a) 1 (b) 3 (c) 5 (d) 8

Ans: (a) 1

32. $(a^4 / a^2) \times a^3 =$
(a) a^4 (b) a^5 (c) a^6 (d) a^8

Ans: (b) a^5

33. The value of $(-1)^{47}$ is
(a) -1 (b) 1 (c) 0 (d) None of these

Ans: (a) -1

34. The multiplicative inverse of -7 is _____
(a) 7 (b) $1/7$ (c) -7 (d) $-1/7$

Ans: (d) $-1/7$

35. The number which is multiplied by $(-8)^{-1}$ to obtain a product equal to 10^{-1} is ____.

(a) $-4/5$ (b) $-3/5$ (c) $-1/5$ (d) $-5/4$

Ans: (a) $-4/5$

36. The multiplicative inverse of $1/5$ is

- (a) 7 (b) 5 (c) 6 (d) 9

Ans: (b) 5

37. Express 256 as a power 4.

- (a) 4^8 (b) 2^8 (c) 4^4 (d) None of these

Ans: (c) 4^4

38. Express 729 as a power of 3

- (a) 3^8 (b) 3^6 (c) 9^3 (d) None of these

Ans: (b) 3^6

39. Express 2048 as a power 2.

- (a) 2^{11} (b) 2^8 (c) 4^3 (d) None of these

Ans: (a) 2^{11}

40. Which one is greater?

- (a) 2^3 (b) 3^2 (c) 1^8 (d) 4^2

Ans: (d) 4^2

41. Express 432 as a product of powers of prime factors.

- (a) $2^3 \times 3^3$ (b) $2^4 \times 3^3$ (c) 16×27 (d) None of these

Ans: (b) $2^4 \times 3^3$

42. The value of $(-1)^{55}$ is

- (a) -1 (b) 1 (c) 0 (d) None of these

Ans: (a) -1

43. The value of $(-1)^{500}$ is

- (a) -1 (b) 1 (c) 0 (d) None of these

Ans: (b) 1

44. The value of 2^8 is

- (a) 128 (b) 256 (c) 512 (d) None of these

Ans: (b) 256

45. Simplify and write in exponential form of $2^2 \times 2^5$

- (a) 2^3 (b) 2^7 (c) 128 (d) None of these

Ans: (b) 2^7

46. Simplify and write in exponential form of $(-4)^{100} \times (-4)^{20}$

- (a) $(-4)^{120}$ (b) $(-4)^{80}$ (c) $(-4)^{2000}$ (d) None of these

Ans: (a) $(-4)^{120}$

47. Simplify and write in exponential form of $5^2 \times 5^7 \times 5^{12}$

- (a) 5^3 (b) 5^7 (c) 5^{21} (d) None of these

Ans: (c) 5^{21}

48. The value of 2^2

- (a) 3 (b) 10 (c) 4 (d) 7

Ans: (c) 4

49. The exponent in the expression 3^7 is _____.

- (a) 1 (b) 7 (c) 0 (d) 3

Ans:

50. The value of 3^0 is _____.

- (a) 0 (b) 3 (c) 1 (d) None of these

Ans: (c) 1

51. Multiplicative inverse of $1/49$ is

- (a) 4 (b) 5 (c) 7^2 (d) -14

Ans: (c) 7^2

52. $a^m / a^n = a^{\dots}$ where m and n are natural numbers.

- (a) $m \cdot n$ (b) $m + n$ (c) $m - n$ (d) $m \div n$

Ans: (c) $m - n$

53. Express $(2a)^4$ in exponential form.

- (a) $4a^3$ (b) $16a^4$ (c) $2a^4$ (d) $8a^4$

Ans: (b) $16a^4$

54. The value of $1/3^2$ is equal to-----

- (a) $1/9$ (b) 1 (c) -6 (d) $1/3$

Ans: (a) $1/9$

55. Find the value of 11^2

- (a) 22 (b) 9 (c) 121 (d) 13

Ans: (c) 121

56. In simplified form $(3^0 + 4^0 + 5^0)^0$ is equal to

- (a) 12 (b) 3 (c) 12 (d) 1

Ans: (d) 1

57. Find the value of $(2/3)^2$

- (a) $4/9$ (b) $9/4$ (c) $-2/9$ (d) 0

Ans: (a) $4/9$

58. In standard form 52, 00, 00,000 is equal to _____.

- (a) 5.2×10^7 (b) 5.2×10^8 (c) 52×10^8 (d) $52 \times 100,00,000$

Ans: (b) 5.2×10^8

59. Usual form of the expression 10^4 is given by _____.

- (a) 100, 00 (b) 1, 0000 (c) 10×10^4 (d) 10,000

Ans: (d) 10,000

60. 1 micron is equals to _____.
(a) $1 / 1000000$ (b) 10^6 m (c) 10^5 m (d) 10^7 m
Ans: (a) $1 / 1000000$

61. The approximate distance of moon from the earth is 384,467,000 m and in exponential form this distance can be written as _____.
(a) $3.84,467 \times 10^8$ m (b) $384,467 \times 10^{-8}$ m
(c) $384,467 \times 10^{-9}$ m (d) $3.844,67 \times 10^{-13}$ m
Ans: (a) $3.84,467 \times 10^8$ m

62. 7×10^{-5} m is the standard form of which of the following _____.
(a) 0.0007 m (b) 0.000007 m (c) 0.0000007 m (d) 0.00007 m
Ans: (d) 0.00007 m

63. The standard form of 4050000 is given by _____.
(a) 4.05×10^6 (b) 40.5×10^9 (c) 405×10^6 (d) 4.05×10^{-6}
Ans: (a) 4.05×10^6

64. Which one of the following is the value of 11^5
(a) 0 (b) 15 (c) 1 (d) None of these
Ans: (d) None of these

65. $(-1)^{\text{Even number}}$ = ----
(a) $2 \times (-1)$ (b) 1 (c) 0 (d) -1^3
Ans: (b) 1

66. $(-1)^{\text{Odd number}}$ = ----
(a) -1 (b) 1 (c) 0 (d) 2
Ans: (a) -1

67. Value of $(3^0 + 2^0) \times 5^0$ is
(a) 1 (b) 25 (c) 2 (d) 0
Ans: (c) 2

68. The value of 6^2 is _____.
(a) 7 (b) 49 (c) 36 (d) 14
Ans: (c) 36

69. The base in the expression of 8^{10} is
(a) 10 (b) 2 (c) 8 (d) 800
Ans: (c) 8

70. The value of 100^0 is _____.
(a) 0 (b) 100 (c) 1 (d) None of these
Ans: (c) 1

71. Find the number from the following expanded form : $9 \times 10^5 + 2 \times 10^2 + 3 \times 10^1$

(a) 900203 (b) 912351 (c) 905302 (d) 900230

Ans: (d) 900230

72. Value of $(2^3)^2$ is given by _____
(a) 64 (b) 32 (c) 12 (d) None of these

Ans: (a) 64

73. The value of $7^2 \div 7^3$ is given by _____.
(a) $1/7$ (b) 7 (c) $1/14$ (d) -7

Ans: (a) $1/7$

74. The value of $1/5^2$ is equal to -----.
(a) -5 (b) 25 (c) -15 (d) $1/25$

Ans: (d) $1/25$

75. In exponential form 140,000,000,000 Kg is given by _____.
(a) 1.4×10^{10} Kg (b) 1.4×10^9 Kg (c) 14×10^8 Kg (d) 1.4×10^{11} Kg

Ans: (d) 1.4×10^{11} Kg

76. The expression, $(5^2 + 7^2 + 3^2)^0$ is equal to
(a) 15^6 (b) -6 (c) 1 (d) 8^3

Ans: (c) 1

77. The value of $(1/6)^2$ is -----
(a) $1/12$ (b) $2/3$ (c) $1/36$ (d) 2

Ans: (c) $1/36$

78. In standard form 56700000 is written as _____.
(a) 5.67×10^7 (b) 567×10^7 (c) 5.67×10^5 (d) 567×100000

Ans: (a) 5.67×10^7

79. Usual form of the expression 9×10^{-5} is given by _____.
(a) 0.00009 (b) 0.000009 (c) 90×10^{-4} (d) 0.09×10^{-3}

Ans: (a) 0.00009

80. The number 86,800,000,000,000,000,000,000 Kg is equal to

(a) 8.68×10^{25} Kg (b) 868×10^{23} Kg (c) 86.8×10^{-25} Kg (d) 868×10^{-23}

Ans: (a) 8.68×10^{25} Kg

81. Charge of an electron is 0.000,000,000,000,000,000,16 coulomb and in exponential form it can be written as _____.

(a) 16×10^{-18} coulomb (b) 1.6×10^{-21} coulomb
(c) 1.6×10^{-19} coulomb (d) 16×10^{-21} coulomb

Ans: (c) 1.6×10^{-19} coulomb

82. 13×10^{-7} Km is the standard form of which of the following _____.

(a) 0.000000013 Km (b) 0.0000013 Km

(c) 0.000000000013 Km (d) 0.00000000013 Km

Ans: (a) 0.000000013 Km

83. The standard form of 9,030,000,000 is given by _____.

(a) 9.03×10^6 (b) 90.3×10^7 (c) 9.03×10^9 (d) 9.03×10^{-9}

Ans: (c) 9.03×10^9

84. Which one of the following is the value of 3^5

(a) 3 (b) 15 (c) 2 (d) 243

Ans: (d) 243

85. Find the value of $5^0 \times 7^0 \times 3^0$

(a) 1 (b) $1/24$ (c) 6 (d) $(1/5) \times 7 \times 3$

Ans: (a) 1

86. 64 in exponential form is _____.

(a) 2^6 (b) 16^2 (c) 2^8 (d) 2^4

Ans: (a) 2^6

87. The value of $2^0 \times 3^0 \times 4^0$ is

(a) 1 (b) 0 (c) 24 (d) None of these

Ans: (a) 1

88. 1024 in exponential form is _____.

(a) 2^6 (b) 16^2 (c) $8(1/2)$ (d) None of these

Ans: (d) None of these

89. The value of $2^2 / 3^3$ in the exponential form is-----.

(a) $(2/3)^5$ (b) $(2/3)^{-1}$ (c) $4/27$ (d) None of these

Ans: (c) $4/27$

90. Find the correct symbol between the numbers 2^3 --- 3^2 .

(a) < (b) > (c) = (d) None of these

Ans: (b) >

91. Find the correct symbol between the numbers 8^2 --- 2^8 .

(a) < (b) > (c) = (d) None of these

Ans: (a) <

92. Simplify the term $(-4)^3$

(a) 64 (b) - 64 (c) 16 (d) None of these

Ans: (b) - 64

93. Simplify the term $(-2)^3 \times (-10)^3$

(a) -80 (b) 8000 (c) 20^8 (d) None of these

Ans: (b) 8000

94. Simplify the term $(-4)^{100} \times (-4)^{20}$

(a) -16 (b) $(4)^{120}$ (c) 16 (d) None of these

Ans: (c) 16

95. Simplify $p^2 \times p^3$
(a) p^6 (b) p^5 (c) p^0 (d) None of these

Ans: (b) p^5

96. Simplify the term $10^5 \div 10^2$
(a) 10^3 (b) 10^5 (c) 10^7 (d) 10^2

Ans: (a) 10^3

97. What is the reciprocal of 3^{-4} ?
(a) 3^4 (b) -3^4 (c) $1/3^4$ (d) Both a & c

Ans: (c) $1/3^4$

98. Which of the following is equal to 10^{-3} ?
(a) 0.003 (b) 0.03 (c) 0.001 (d) 3

Ans: (c) 0.001

99. The standard form of 0.00001275 is
(a) 1.275×10^{-5} (b) 1.275×10^5 (c) 12.75×10^{-6} (d) 127.5×10^{-7}

Ans: (a) 1.275×10^{-5}

100. The standard form of 7,200,000
(a) 7.2×10^5 (b) 7.2×10^6 (c) 72×10^5 (d) 72×10^6

Ans: (b) 7.2×10^6